

Logistic Regression

Let $y_i \sim \text{Bernoulli}(\pi_i)$ for $i = 1, \dots, n$. The logistic regression model is defined by the log-odds (logit) link:

$$\ln \left(\frac{\pi_i}{1 - \pi_i} \right) = \eta_i = \sum_{j=1}^p \beta_j x_{ij}$$

This linear component η_i can be written in vector form as $\eta_i = \mathbf{x}'_i \boldsymbol{\beta}$.

The log-likelihood function $l(\boldsymbol{\beta})$ is derived from the product of the Bernoulli probability mass functions:

$$\begin{aligned} L(\boldsymbol{\beta}|\mathbf{y}) &= \prod_{i=1}^n \pi_i^{y_i} (1 - \pi_i)^{1-y_i} \\ l(\boldsymbol{\beta}|\mathbf{y}) &= \ln L(\boldsymbol{\beta}|\mathbf{y}) \\ &= \sum_{i=1}^n [y_i \ln(\pi_i) + (1 - y_i) \ln(1 - \pi_i)] \end{aligned}$$

We can express π_i and $1 - \pi_i$ in terms of η_i :

$$\pi_i = \frac{e^{\eta_i}}{1 + e^{\eta_i}} \quad \text{and} \quad 1 - \pi_i = \frac{1}{1 + e^{\eta_i}}$$

Maximum Likelihood Estimation (MLE)

To find the MLE, we must solve the score equations by setting the first partial derivatives of the log-likelihood to zero.

First, find the partial derivative of π_i with respect to a single parameter β_j :

$$\frac{\partial \pi_i}{\partial \beta_j} = \frac{\partial \pi_i}{\partial \eta_i} \frac{\partial \eta_i}{\partial \beta_j} = \left[\frac{e^{\eta_i}(1 + e^{\eta_i}) - e^{\eta_i}(e^{\eta_i})}{(1 + e^{\eta_i})^2} \right] x_{ij} = \frac{e^{\eta_i}}{(1 + e^{\eta_i})^2} x_{ij} = \pi_i(1 - \pi_i)x_{ij}$$

Now, find the partial derivative of $l(\boldsymbol{\beta})$ with respect to β_j :

$$\begin{aligned}
\frac{\partial l}{\partial \beta_j} &= \sum_{i=1}^n \left[\frac{y_i}{\pi_i} \frac{\partial \pi_i}{\partial \beta_j} + \frac{1 - y_i}{1 - \pi_i} \left(-\frac{\partial \pi_i}{\partial \beta_j} \right) \right] \\
&= \sum_{i=1}^n \left[\frac{y_i}{\pi_i} - \frac{1 - y_i}{1 - \pi_i} \right] \frac{\partial \pi_i}{\partial \beta_j} \\
&= \sum_{i=1}^n \left[\frac{y_i(1 - \pi_i) - (1 - y_i)\pi_i}{\pi_i(1 - \pi_i)} \right] (\pi_i(1 - \pi_i)x_{ij}) \\
&= \sum_{i=1}^n [y_i - y_i\pi_i - \pi_i + y_i\pi_i] x_{ij} \\
&= \sum_{i=1}^n (y_i - \pi_i)x_{ij}
\end{aligned}$$

To find the MLE, we set this to zero for all $j = 1, \dots, p$:

$$\frac{\partial l}{\partial \beta_j} = \sum_{i=1}^n (y_i - \pi_i)x_{ij} = 0$$

This is a non-linear system of equations in $(\beta_1, \dots, \beta_p)$. Analytically, we cannot solve this.

Newton-Raphson Method

We need an iterative method to find the solution. Let $\mathbf{u}(\boldsymbol{\beta})$ be the score vector (gradient):

$$\mathbf{u}(\boldsymbol{\beta}) = \nabla l(\boldsymbol{\beta}) = \left(\frac{\partial l}{\partial \beta_j} \right)_{j=1}^p$$

We want to find $\hat{\boldsymbol{\beta}}$ such that $\mathbf{u}(\hat{\boldsymbol{\beta}}) = \mathbf{0}$.

Recall the 1D Taylor expansion around $x^{(t)}$:

$$f(x) \approx f(x^{(t)}) + f'(x^{(t)})(x - x^{(t)})$$

Setting $f(x) = 0$ and solving for x gives the update rule $x^{(t+1)} = x^{(t)} - \frac{f(x^{(t)})}{f'(x^{(t)})}$.

For the multivariate case, we use a Taylor expansion on the score vector $\mathbf{u}(\boldsymbol{\beta})$ around the current estimate $\boldsymbol{\beta}^{(t)}$:

$$\mathbf{u}(\boldsymbol{\beta}) \approx \mathbf{u}(\boldsymbol{\beta}^{(t)}) + H(\boldsymbol{\beta}^{(t)})(\boldsymbol{\beta} - \boldsymbol{\beta}^{(t)})$$

where $H(\boldsymbol{\beta})$ is the Hessian matrix (matrix of second partial derivatives):

$$H(\boldsymbol{\beta}) = \left(\frac{\partial^2 l}{\partial \beta_j \partial \beta_k} \right)$$

We set $\mathbf{u}(\boldsymbol{\beta}) = \mathbf{0}$ at the next step, $\boldsymbol{\beta} = \boldsymbol{\beta}^{(t+1)}$:

$$\begin{aligned}\mathbf{0} &\approx \mathbf{u}^{(t)} + H^{(t)}(\boldsymbol{\beta}^{(t+1)} - \boldsymbol{\beta}^{(t)}) \\ -H^{(t)}(\boldsymbol{\beta}^{(t+1)} - \boldsymbol{\beta}^{(t)}) &= \mathbf{u}^{(t)} \\ \boldsymbol{\beta}^{(t+1)} - \boldsymbol{\beta}^{(t)} &= -[H^{(t)}]^{-1}\mathbf{u}^{(t)} \\ \boldsymbol{\beta}^{(t+1)} &= \boldsymbol{\beta}^{(t)} - [H^{(t)}]^{-1}\mathbf{u}^{(t)}\end{aligned}$$

This is the Newton-Raphson algorithm.

Stopping Conditions

We stop the iterations when:

- (i) A fixed number of recursions is reached.
- (ii) The change in parameters is small: $\|\boldsymbol{\beta}^{(t+1)} - \boldsymbol{\beta}^{(t)}\| < \epsilon$ for some predefined constant ϵ .
- (iii) The gradient (score) is close to zero: $\|\mathbf{u}^{(t)}\| < \delta$.

A potential issue is the existence of multiple local maxima of the log-likelihood.

Problem: At every step of the recursion, a matrix inversion is involved.

Fisher Scoring

Fisher Scoring replaces the observed Hessian matrix $H^{(t)}$ with its expected value, which is related to the Fisher Information matrix. Let $J = E[H(\boldsymbol{\beta})]$. (Note: The Fisher Information is standardly $I(\boldsymbol{\beta}) = -E[H(\boldsymbol{\beta})]$). The update rule becomes:

$$\boldsymbol{\beta}^{(t+1)} = \boldsymbol{\beta}^{(t)} - [J^{(t)}]^{-1}\mathbf{u}^{(t)}$$

This is an approximation to Newton-Raphson and may take longer to converge. We get the variance of the MLE from the converged value of the Fisher information matrix: $\text{Var}(\hat{\boldsymbol{\beta}}) \approx [I(\hat{\boldsymbol{\beta}})]^{-1}$ or $\text{Var}(\hat{\boldsymbol{\beta}}) \approx [-J(\hat{\boldsymbol{\beta}})]^{-1}$.

Generalized Linear Model (GLM)

Considers a response variable Y_i from a natural exponential family:

$$f(y_i; \theta_i, \phi) = a(\theta).b(y_i)\exp(y_i.Q(\theta_i))$$

Let $\mu_i = E[Y_i]$. A GLM relates μ_i to a linear predictor $\eta_i = \sum_j \beta_j x_{ij}$ via a link function $g(\cdot)$:

$$g(\mu_i) = \eta_i = \sum_j \beta_j x_{ij}$$

Example: Multiple Linear Regression

$$Y_i = \sum_j \beta_j x_{ij} + \epsilon_i, \quad \epsilon_i \sim N(0, \sigma^2)$$

Here, $E[Y_i] = \mu_i = \sum_j \beta_j x_{ij}$. The link function is the identity link: $g(u) = u$.

Example: Logistic Regression

Here, $E[Y_i] = \pi_i$. So $\mu_i = \pi_i$. The link function is the logit link:

$$g(\mu_i) = g(\pi_i) = \ln \left(\frac{\pi_i}{1 - \pi_i} \right) = \sum_j \beta_j x_{ij}$$

$$g(u) = \ln \left(\frac{u}{1 - u} \right)$$